

CHAPTER 11

Reactions of Alkyl Halides: Nucleophilic Substitutions and Eliminations



FIGURE 11.1 Competition occurs throughout nature. In chemistry, competition often occurs between alternative reaction pathways, such as in the substitution and elimination reactions of alkyl halides. (credit: modification of work “Bull moose fight” by Grand Teton, National Parks Service/Flickr, Public Domain)

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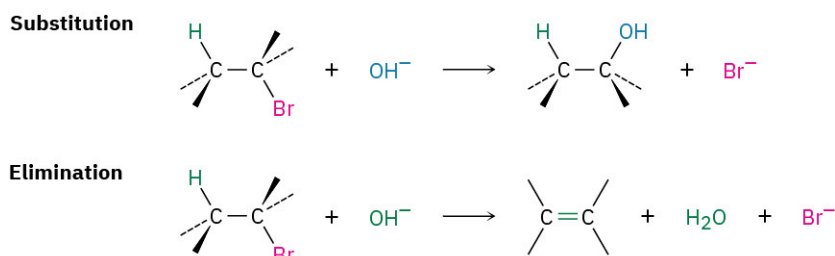
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WHY THIS CHAPTER? Nucleophilic substitution and base-induced elimination are two of the most widely occurring and versatile reactions in organic chemistry, both in the laboratory and in biological pathways. We'll

look at them closely in this chapter to see how they occur, what their characteristics are, and how they can be used. We'll begin with substitution reactions.

We saw in the preceding chapter that the carbon–halogen bond in an alkyl halide is polar and that the carbon atom is electron-poor. Thus, alkyl halides are electrophiles, and much of their chemistry involves polar reactions with nucleophiles and bases. Alkyl halides do one of two things when they react with a nucleophile/base such as hydroxide ion: they either undergo *substitution* of the X group by the nucleophile, or they undergo *elimination* of HX to yield an alkene.



11.1 The Discovery of Nucleophilic Substitution Reactions

Discovery of the nucleophilic substitution reaction of alkyl halides dates back to work carried out by the German chemist Paul Walden in 1896. Walden found that the pure enantiomeric (+)- and (–)-malic acids could be interconverted through a series of simple substitution reactions. When Walden treated (–)-malic acid with PCl_5 , he isolated (+)-chlorosuccinic acid. This, on treatment with wet Ag_2O , gave (+)-malic acid. Similarly, reaction of (+)-malic acid with PCl_5 gave (–)-chlorosuccinic acid, which was converted into (–)-malic acid when treated with wet Ag_2O . The full cycle of reactions is shown in **FIGURE 11.2**.

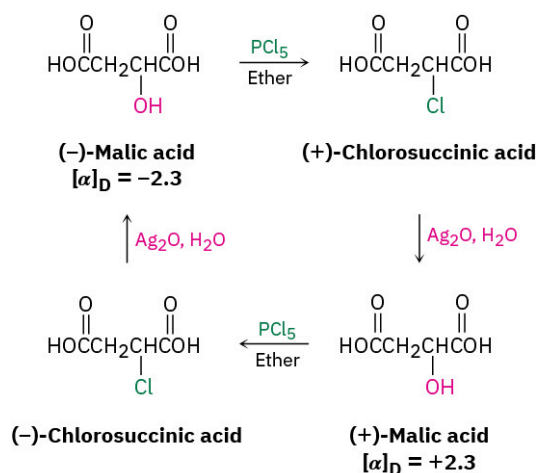


FIGURE 11.2 Walden's cycle of reactions interconverting (+)- and (–)-malic acids.

At the time, the results were astonishing. The eminent chemist Emil Fischer called Walden's discovery “the most remarkable observation made in the field of optical activity since the fundamental observations of Pasteur.” Because (–)-malic acid was converted into (+)-malic acid, some reactions in the cycle must have occurred with a change, or inversion, of configuration at the chirality center. But which ones, and how? (Remember from **Section 5.5** that the direction of light rotation and the configuration of a chirality center aren't directly related. You can't tell by looking at the sign of rotation whether a change in configuration has occurred during a reaction.)

Today, we refer to the transformations taking place in Walden's cycle as **nucleophilic substitution reactions** because each step involves the substitution of one nucleophile (chloride ion, Cl^- , or hydroxide ion, HO^-) by another. Nucleophilic substitution reactions are one of the most common and versatile reaction types in organic chemistry.



Following the work of Walden, further investigations were undertaken during the 1920s and 1930s to clarify the

mechanism of nucleophilic substitution reactions and to find out how inversions of configuration occur. Among the first series studied was one that interconverted the two enantiomers of 1-phenyl-2-propanol (**FIGURE 11.3**). Although this particular series of reactions involves nucleophilic substitution of an alkyl *para*-toluenesulfonate (called a **tosylate**) rather than an alkyl halide, exactly the same type of reaction is involved as that studied by Walden. For all practical purposes, the entire tosylate group acts as if it were simply a halogen substituent. (In fact, when you see a tosylate substituent in a molecule, do a mental substitution and tell yourself that you're dealing with an alkyl halide.)

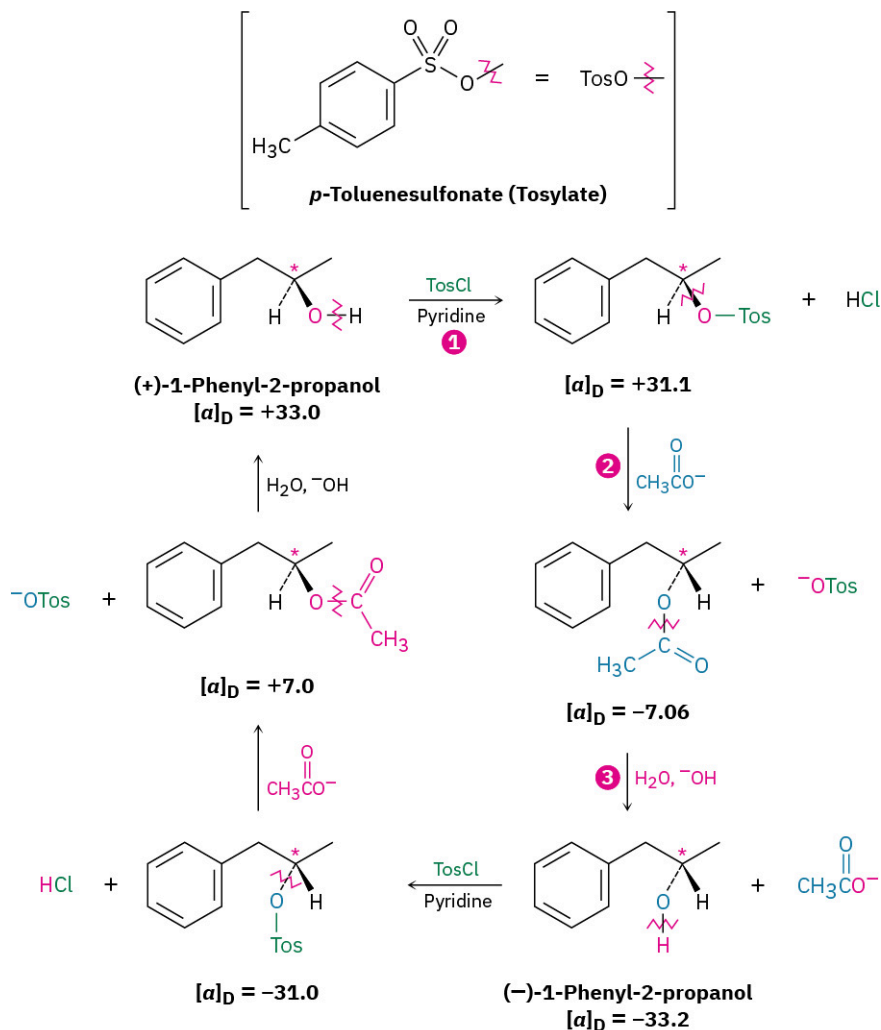
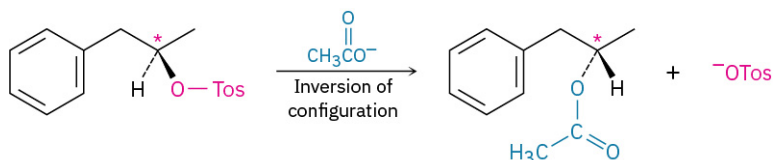


FIGURE 11.3 A Walden cycle interconverting (+) and (-) enantiomers of 1-phenyl-2-propanol. Chirality centers are marked by asterisks, and the bonds broken in each reaction are indicated by red wavy lines. The inversion of chirality occurs in step 2, where acetate ion substitutes for tosylate ion.

In the three-step reaction sequence shown in **FIGURE 11.3**, (+)-1-phenyl-2-propanol is interconverted with its (-) enantiomer, so at least one of the three steps must involve an inversion of configuration at the chirality center. Step 1, formation of a tosylate, occurs by breaking the O-H bond of the alcohol rather than the C-O bond to the chiral carbon, so the configuration around the carbon is unchanged. Similarly, step 3, hydroxide-ion cleavage of the acetate, takes place without breaking the C-O bond at the chirality center. Thus, the inversion of stereochemical configuration must take place in step 2, the nucleophilic substitution of tosylate ion by acetate ion.



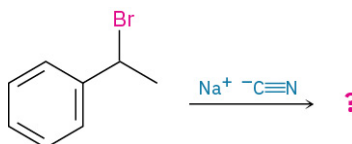
From this and nearly a dozen other series of similar reactions, researchers concluded that the nucleophilic substitution reaction of a primary or secondary alkyl halide or tosylate always proceeds with inversion of configuration. (Tertiary alkyl halides and tosylates, as we'll see shortly, give different stereochemical results and react by a different mechanism than the primary and secondary ones.)



WORKED EXAMPLE 11.1

Predicting the Stereochemistry of a Nucleophilic Substitution Reaction

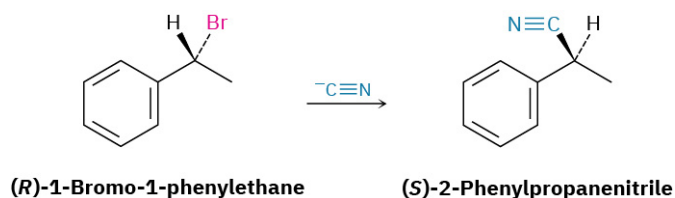
What product would you expect from a nucleophilic substitution reaction of (*R*)-1-bromo-1-phenylethane with cyanide ion, $\text{C}\equiv\text{N}^-$, as nucleophile? Show the stereochemistry of both reactant and product, assuming that inversion of configuration occurs.



Strategy

Draw the *R* enantiomer of the reactant, and then change the configuration of the chirality center while replacing the Br^- with a CN^- .

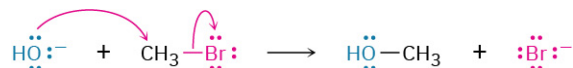
Solution



PROBLEM 11-1 What product would you expect from a nucleophilic substitution reaction of (*S*)-2-bromohexane with acetate ion, CH_3CO_2^- ? Assume that inversion of configuration occurs, and show the stereochemistry of both the reactant and product.

11.2 The $\text{S}_\text{N}2$ Reaction

In almost all chemical reactions, there is a direct relationship between the rate at which the reaction occurs and the concentrations of the reactants. When we measure this relationship, we measure the **kinetics** of the reaction. For example, let's look at the kinetics of a simple nucleophilic substitution—the reaction of CH_3Br with OH^- to yield CH_3OH plus Br^- .



With a given temperature, solvent, and concentration of reactants, the substitution occurs at a certain rate. If we double the concentration of OH^- , the frequency of encounters between reaction partners doubles and we find that the reaction rate also doubles. Similarly, if we double the concentration of CH_3Br , the reaction rate again doubles. We call such a reaction, in which the rate is linearly dependent on the concentrations of two species, a **second-order reaction**. Mathematically, we can express this second-order dependence of the nucleophilic substitution reaction by setting up a **rate equation**. As either $[\text{RX}]$ or $[\text{OH}^-]$ changes, the rate of the reaction changes proportionately.

$$\begin{aligned}\text{Reaction rate} &= \text{Rate of disappearance of reactant} \\ &= k \times [\text{RX}] \times [\text{OH}^-]\end{aligned}$$

where

[RX] = CH₃Br concentration in molarity

[[−]OH] = [−]OH concentration in molarity

k = a constant value (the rate constant)

A mechanism that accounts for both the inversion of configuration and the second-order kinetics that are observed with nucleophilic substitution reactions was suggested in 1937 by the British chemists E. D. Hughes and Christopher Ingold, who formulated what they called the **S_N2 reaction**—short for *substitution, nucleophilic, bimolecular*. (**Bimolecular** means that two molecules, nucleophile and alkyl halide, take part in the step whose kinetics are measured.)

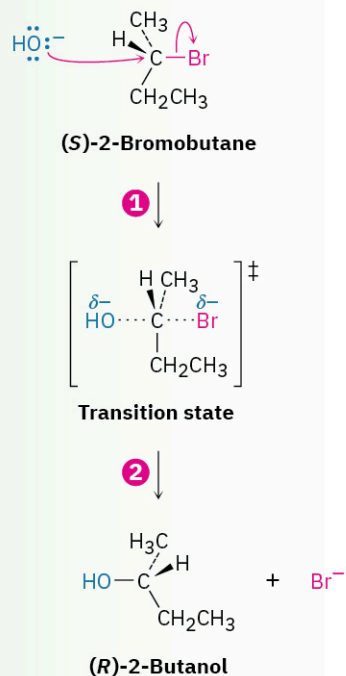
The essential feature of the S_N2 mechanism is that it takes place in a single step, without intermediates, when the incoming nucleophile reacts with the alkyl halide or tosylate (the *substrate*) from a direction opposite the group that is displaced (the *leaving group*). As the nucleophile comes in on one side of the substrate and bonds to the carbon, the halide or tosylate departs from the other side, thereby inverting the stereochemical configuration. The process is shown in **FIGURE 11.4** for the reaction of (*S*)-2-bromobutane with HO[−] to give (*R*)-2-butanol.

FIGURE 11.4 MECHANISM

The mechanism of the S_N2 reaction. The reaction takes place in a single step when the incoming nucleophile approaches from a direction 180° away from the leaving halide ion, thereby inverting the stereochemistry at carbon.

- 1 The nucleophile [−]OH uses its lone-pair electrons to attack the alkyl halide carbon 180° away from the departing halogen. This leads to a transition state with a partially formed C–OH bond and a partially broken C–Br bond.

- 2 The stereochemistry at carbon is inverted as the C–OH bond forms fully and the bromide ion departs with the electron pair from the former C–Br bond.



As shown in **FIGURE 11.4**, the S_N2 reaction occurs when an electron pair on the nucleophile Nu:[−] forces out the group X:[−], which takes with it the electron pair from the former C–X bond. This occurs through a transition state in which the new Nu–C bond is partially formed at the same time that the old C–X bond is partially broken and in which the negative charge is shared by both the incoming nucleophile and the outgoing halide ion. The transition state for this inversion has the remaining three bonds to carbon in a planar arrangement (**FIGURE 11.5**).

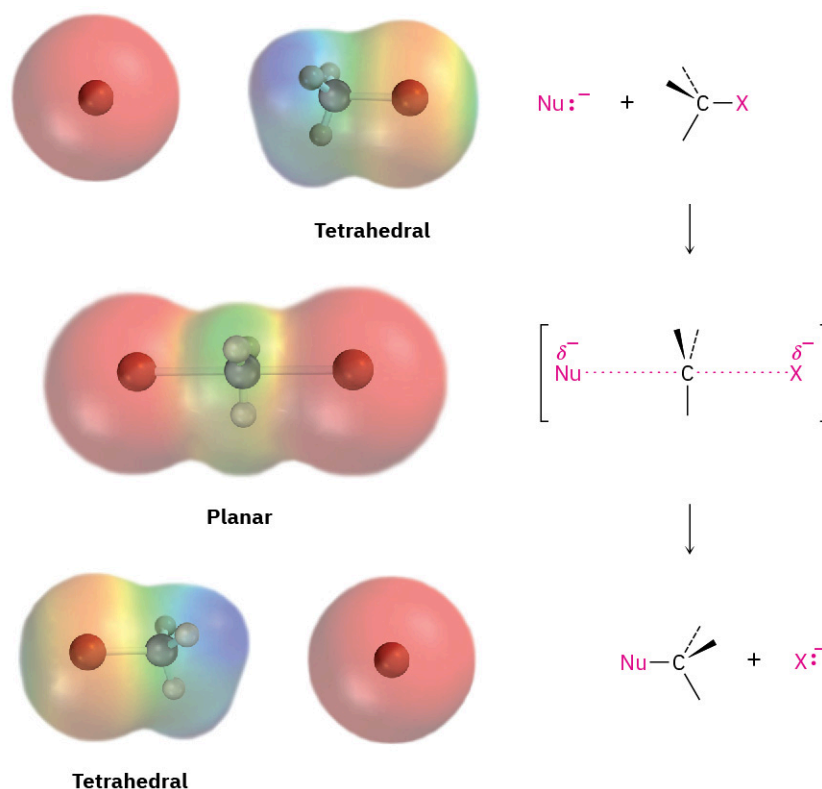
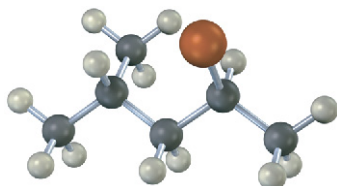


FIGURE 11.5 The transition state of an S_N2 reaction has a planar arrangement of the carbon atom and the remaining three groups. Electrostatic potential maps show that **negative charge** is delocalized in the transition state.

The mechanism proposed by Hughes and Ingold is fully consistent with experimental results, explaining both stereochemical and kinetic data. Thus, the requirement for a backside approach of the entering nucleophile (180° away from the departing X group) causes the stereochemistry of the substrate to invert, much like an umbrella turning inside-out in the wind. The Hughes–Ingold mechanism also explains why second-order kinetics are observed: the S_N2 reaction occurs in a single step that involves both alkyl halide and nucleophile. Two molecules are involved in the step whose rate is measured.

PROBLEM What product would you expect to obtain from S_N2 reaction of OH^- with (*R*)-2-bromobutane? Show **11-2** the stereochemistry of both the reactant and product.

PROBLEM Assign configuration to the following substance, and draw the structure of the product that would **11-3** result from nucleophilic substitution reaction with HS^- (reddish brown = Br):



11.3 Characteristics of the S_N2 Reaction

Now that we know how S_N2 reactions occur, we need to see how they can be used and what variables affect them. Some S_N2 reactions are fast, and some are slow; some take place in high yield and others in low yield. Understanding the factors involved can be of tremendous value. Let's begin by recalling a few things about reaction rates in general.

The rate of a chemical reaction is determined by the activation energy $\Delta G^\ddagger$, the energy difference between reactant ground state and transition state. A change in reaction conditions can affect $\Delta G^\ddagger$ either by changing the reactant energy level or by changing the transition-state energy level. Lowering the reactant energy or raising the transition-state energy increases $\Delta G^\ddagger$ and decreases the reaction rate; raising the reactant energy or

decreasing the transition-state energy decreases $\Delta G^\ddagger$ and increases the reaction rate (FIGURE 11.6). We'll see examples of all these effects as we look at S_N2 reaction variables.

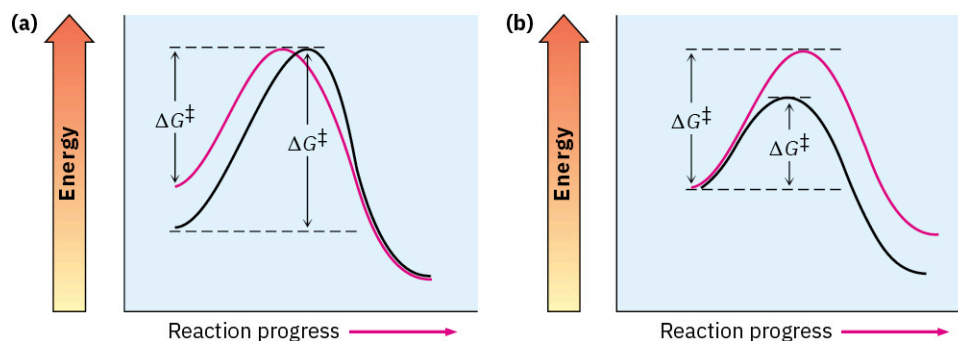


FIGURE 11.6 The effects of changes in reactant and transition-state energy levels on reaction rate. (a) A higher reactant energy level (red curve) corresponds to a faster reaction (smaller $\Delta G^\ddagger$). (b) A higher transition-state energy level (red curve) corresponds to a slower reaction (larger $\Delta G^\ddagger$).

Steric Effects in the S_N2 Reaction

The first S_N2 reaction variable to look at is the structure of the substrate. Because the S_N2 transition state involves partial bond formation between the incoming nucleophile and the alkyl halide carbon atom, it seems reasonable that a hindered, bulky substrate should prevent easy approach of the nucleophile, making bond formation difficult. In other words, the transition state for reaction of a sterically hindered substrate, whose carbon atom is “shielded” from the approach of the incoming nucleophile, is higher in energy and forms more slowly than the corresponding transition state for a less hindered substrate (FIGURE 11.7).

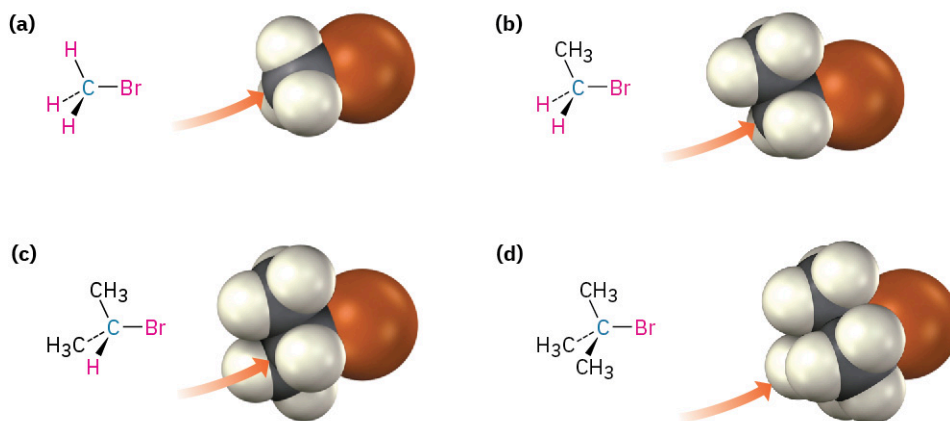
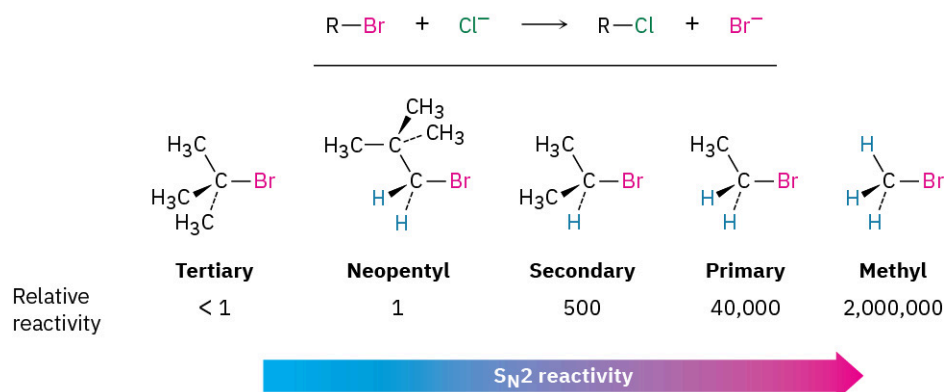
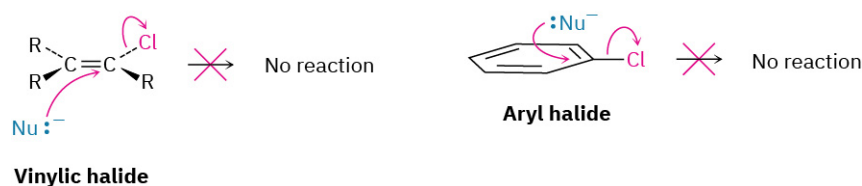


FIGURE 11.7 Steric hindrance to the S_N2 reaction. As the models indicate, the carbon atom in (a) bromomethane is readily accessible, resulting in a fast S_N2 reaction. The carbon atoms in (b) bromoethane (primary), (c) 2-bromopropane (secondary), and (d) 2-bromo-2-methylpropane (tertiary) are successively more hindered, resulting in successively slower S_N2 reactions.

As FIGURE 11.7 shows, the difficulty of nucleophile approach increases as the three substituents bonded to the halo-substituted carbon atom increase in size. Methyl halides are by far the most reactive substrates in S_N2 reactions, followed by primary alkyl halides such as ethyl and propyl. Alkyl branching at the reacting center, as in isopropyl halides (2°), slows the reaction greatly, and further branching, as in *tert*-butyl halides (3°), effectively halts the reaction. Even branching one carbon away from the reacting center, as in 2,2-dimethylpropyl (*neopentyl*) halides, greatly hinders nucleophilic displacement. As a result, S_N2 reactions occur only at relatively unhindered sites and are normally useful only with methyl halides, primary halides, and a few simple secondary halides. Relative reactivities for some different substrates are as follows:

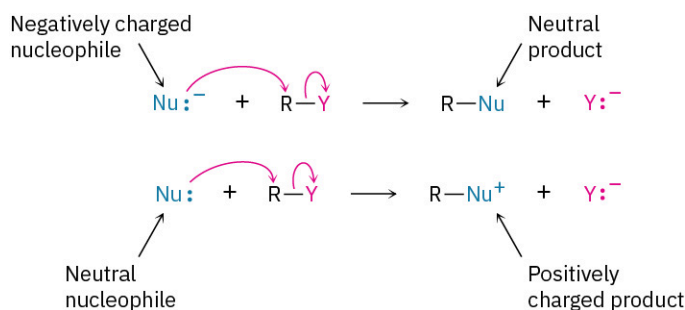


Vinyl halides ($\text{R}_2\text{C}=\text{CRX}$) and aryl halides are not shown on this reactivity list because they are unreactive toward $\text{S}_{\text{N}}2$ displacement. This lack of reactivity is due to steric factors: the incoming nucleophile would have to approach in the plane of the carbon-carbon double bond and burrow through part of the molecule to carry out a backside displacement.

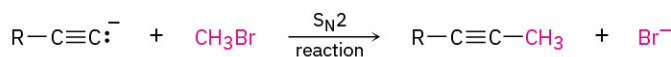


The Nucleophile

Another variable that has a major effect on the $\text{S}_{\text{N}}2$ reaction is the nature of the nucleophile. Any species, either neutral or negatively charged, can act as a nucleophile as long as it has an unshared pair of electrons; that is, as long as it is a Lewis base. If the nucleophile is negatively charged, the product is neutral; if the nucleophile is neutral, the product is positively charged.



A wide array of substances can be prepared using nucleophilic substitution reactions. In fact, we've already seen examples in previous chapters. For instance, the reaction of an acetylide anion with an alkyl halide, discussed in [Section 9.8](#), is an $\text{S}_{\text{N}}2$ reaction in which the acetylide nucleophile displaces a halide leaving group.



An acetylide anion

TABLE 11.1 lists some nucleophiles in the order of their reactivity, shows the products of their reactions with bromomethane, and gives the relative rates of their reactions. There are large differences in the rates at which various nucleophiles react.

TABLE 11.1 Some S_N2 Reactions with Bromomethane

$\text{Nu}^- + \text{CH}_3\text{Br} \rightarrow \text{CH}_3\text{Nu} + \text{Br}^-$				
Nucleophile		Product		Relative rate of reaction
Formula	Name	Formula	Name	
H ₂ O	Water	CH ₃ OH ₂ ⁺	Methylhydronium ion	1
CH ₃ CO ₂ ⁻	Acetate	CH ₃ CO ₂ CH ₃	Methyl acetate	500
NH ₃	Ammonia	CH ₃ NH ₃ ⁺	Methylammonium ion	700
Cl ⁻	Chloride	CH ₃ Cl	Chloromethane	1,000
HO ⁻	Hydroxide	CH ₃ OH	Methanol	10,000
CH ₃ O ⁻	Methoxide	CH ₃ OCH ₃	Dimethyl ether	25,000
I ⁻	Iodide	CH ₃ I	Iodomethane	100,000
⁻ CN	Cyanide	CH ₃ CN	Acetonitrile	125,000
HS ⁻	Hydrosulfide	CH ₃ SH	Methanethiol	125,000

What are the reasons for the reactivity differences observed in TABLE 11.1? Why do some reactants appear to be much more “nucleophilic” than others? The answers to these questions aren’t straightforward. Part of the problem is that the term *nucleophilicity* is imprecise. The term is usually taken to be a measure of the affinity of a nucleophile for a carbon atom in the S_N2 reaction, but the reactivity of a given nucleophile can change from one reaction to the next. The exact nucleophilicity of a species in a given reaction depends on the substrate, the solvent, and even the reactant concentrations. Detailed explanations for the observed nucleophilicities aren’t always simple, but some trends can be detected from the data of TABLE 11.1.

- **Nucleophilicity roughly parallels basicity** when comparing nucleophiles that have the same reacting atom. Thus, OH⁻ is both more basic and more nucleophilic than acetate ion, CH₃CO₂⁻, which in turn is more basic and more nucleophilic than H₂O. Since “nucleophilicity” is usually taken as the affinity of a Lewis base for a carbon atom in the S_N2 reaction and “basicity” is the affinity of a base for a proton, it’s easy to see why there might be a correlation between the two kinds of behavior.
- **Nucleophilicity usually increases going down a column of the periodic table.** Thus, HS⁻ is more nucleophilic than HO⁻, and the halide reactivity order is I⁻ > Br⁻ > Cl⁻. Going down the periodic table, elements have their valence electrons in successively larger shells where they are successively farther from the nucleus, less tightly held, and consequently more reactive. This matter is complex, though, and the nucleophilicity order can change depending on the solvent.
- **Negatively charged nucleophiles are usually more reactive than neutral ones.** As a result, S_N2 reactions are often carried out under basic conditions rather than neutral or acidic conditions.

PROBLEM What product would you expect from S_N2 reaction of 1-bromobutane with each of the following?

11-4 (a) NaI (b) KOH (c) H–C≡C–Li (d) NH₃

PROBLEM Which substance in each of the following pairs is more reactive as a nucleophile? Explain.

11-5 (a) (CH₃)₂N⁻ or (CH₃)₂NH (b) (CH₃)₃B or (CH₃)₃N (c) H₂O or H₂S

The Leaving Group

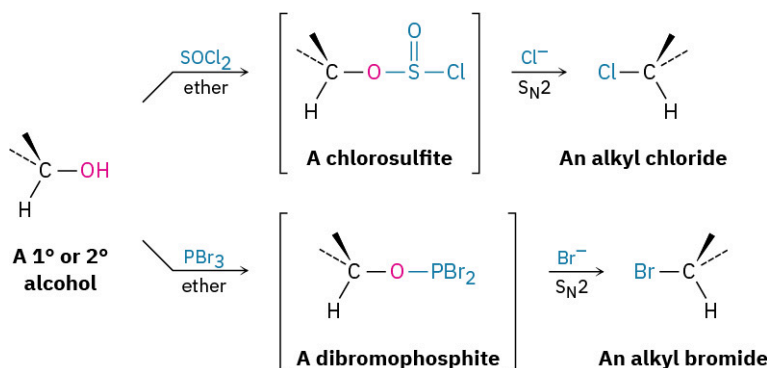
Still another variable that can affect the S_N2 reaction is the nature of the group displaced by the incoming

nucleophile, the **leaving group**. Because the leaving group is expelled with a negative charge in most S_N2 reactions, the best leaving groups are those that best stabilize the negative charge in the transition state. The greater the extent of charge stabilization by the leaving group, the lower the energy of the transition state and the more rapid the reaction. But as we saw in [Section 2.8](#), the groups that best stabilize a negative charge are also the weakest bases. Thus, weak bases such as Cl^- , Br^- , and tosylate ion make good leaving groups, while strong bases such as OH^- and NH_2^- make poor leaving groups.

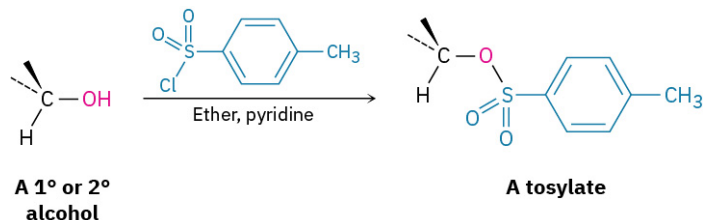
Relative reactivity	$\text{OH}^-, \text{NH}_2^-, \text{OR}^-$	F^-	Cl^-	Br^-	I^-	TosO^-
	$\ll 1$	1	200	10,000	30,000	60,000



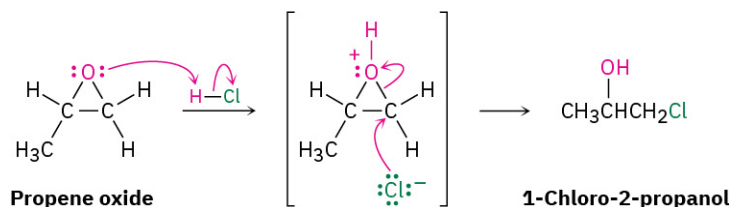
It's just as important to know which are poor leaving groups as to know which are good, and the preceding data clearly indicate that F^- , HO^- , RO^- , and H_2N^- are not displaced by nucleophiles. In other words, alkyl fluorides, alcohols, ethers, and amines do not typically undergo S_N2 reactions. To carry out an S_N2 reaction with an alcohol, it's necessary to convert the OH^- into a better leaving group. This, in fact, is just what happens when a primary or secondary alcohol is converted into either an alkyl chloride by reaction with SOCl_2 or an alkyl bromide by reaction with PBr_3 ([Section 10.5](#)).



Alternatively, an alcohol can be made more reactive toward nucleophilic substitution by treating it with *para*-toluenesulfonyl chloride to form a tosylate. As noted previously, tosylates are **even** more reactive than halides in nucleophilic substitutions. Note that tosylate formation does not change the configuration of the oxygen-bearing carbon because the C–O bond is not broken.



The one general exception to the rule that ethers don't typically undergo S_N2 reactions pertains to epoxides, the three-membered cyclic ethers that we saw in [Section 8.7](#). Because of the angle strain in their three-membered ring, epoxides are much more reactive than other ethers. They react with aqueous acid to give 1,2-diols, as we saw in [Section 8.7](#), and they react readily with many other nucleophiles as well. Propene oxide, for instance, reacts with HCl to give 1-chloro-2-propanol by a S_N2 backside attack on the less hindered primary carbon atom. We'll look at the process in more detail in [Section 18.5](#).



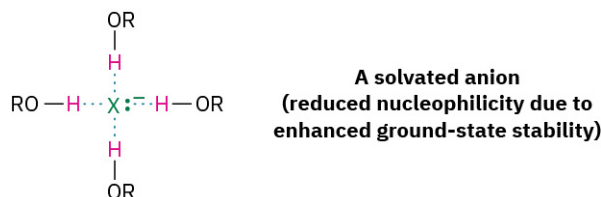
PROBLEM Rank the following compounds in order of their expected reactivity toward S_N2 reaction:

11-6 CH₃Br, CH₃OTos, (CH₃)₃CCl, (CH₃)₂CHCl

The Solvent

The rates of S_N2 reactions are strongly affected by the solvent. Protic solvents—those that contain an –OH or –NH group—are generally the worst for S_N2 reactions, while polar aprotic solvents, which are polar but don't have an –OH or –NH group, are the best.

Protic solvents, such as methanol and ethanol, slow down S_N2 reactions by **solvation** of the reactant nucleophile. The solvent molecules hydrogen-bond to the nucleophile and form a cage around it, thereby lowering its energy and reactivity.



In contrast with protic solvents—which decrease the rates of S_N2 reactions by lowering the ground-state energy of the nucleophile—polar aprotic solvents increase the rates of S_N2 reactions by raising the ground-state energy of the nucleophile. Acetonitrile (CH₃CN), dimethylformamide [(CH₃)₂NCHO, abbreviated DMF], and dimethyl sulfoxide [(CH₃)₂SO, abbreviated DMSO] are particularly useful. A solvent known as hexamethylphosphoramide {(CH₃)₆P=O, abbreviated HMPA} can also be useful but it should only be handled with great care and not be allowed to touch the eyes or skin. These solvents can dissolve many salts because of their high polarity, but they tend to solvate metal cations rather than nucleophilic anions. As a result, the bare, unsolvated anions have a greater nucleophilicity and S_N2 reactions take place at correspondingly increased rates. For instance, a rate increase of 200,000 has been observed on changing from methanol to HMPA for the reaction of azide ion with 1-bromobutane.

	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2-\text{Br} + \text{N}_3^- \longrightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2-\text{N}_3 + \text{Br}^-$					
Solvent	CH ₃ OH	H ₂ O	DMSO	DMF	CH ₃ CN	HMPA
Relative reactivity	1	7	1300	2800	5000	200,000

PROBLEM Organic solvents like benzene, ether, and chloroform are neither protic nor strongly polar. What effect would you expect these solvents to have on the reactivity of a nucleophile in S_N2 reactions?

11-7

A Summary of S_N2 Reaction Characteristics

The effects on S_N2 reactions of the four variables—substrate structure, nucleophile, leaving group, and solvent—are summarized in the following statements and in the energy diagrams of **FIGURE 11.8**:

- Substrate** Steric hindrance raises the energy of the S_N2 transition state, increasing $\Delta G^\ddagger$ and decreasing the reaction rate (**FIGURE 11.8a**). As a result, S_N2 reactions are best for methyl and primary substrates. Secondary substrates react slowly, and tertiary substrates do not react by an S_N2 mechanism.
- Nucleophile** Basic, negatively charged nucleophiles are less stable and have a higher ground-state energy than neutral ones, decreasing $\Delta G^\ddagger$ and increasing the S_N2 reaction rate (**FIGURE 11.8b**).

Leaving group Solvent

Good leaving groups (more stable anions) lower the energy of the transition state, decreasing $\Delta G^\ddagger$ and increasing the S_N2 reaction rate (FIGURE 11.8c).

Protic solvents solvate the nucleophile, thereby lowering its ground-state energy, increasing $\Delta G^\ddagger$, and decreasing the S_N2 reaction rate. Polar aprotic solvents surround the accompanying cation but not the nucleophilic anion, thereby raising the ground-state energy of the nucleophile, decreasing $\Delta G^\ddagger$, and increasing the reaction rate (FIGURE 11.8d).

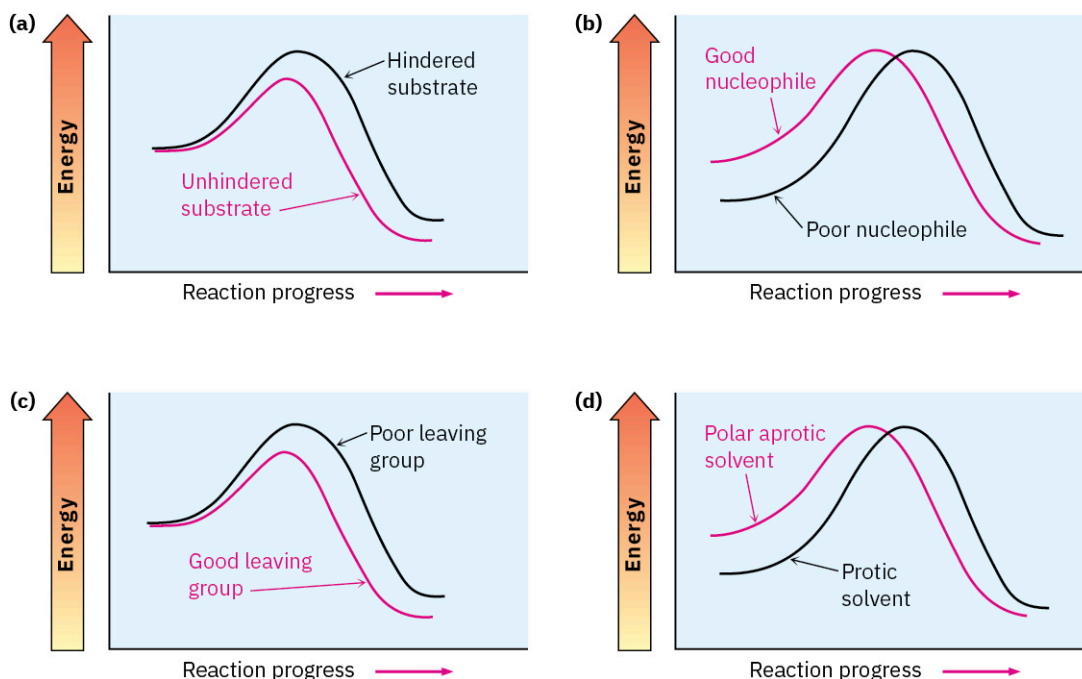
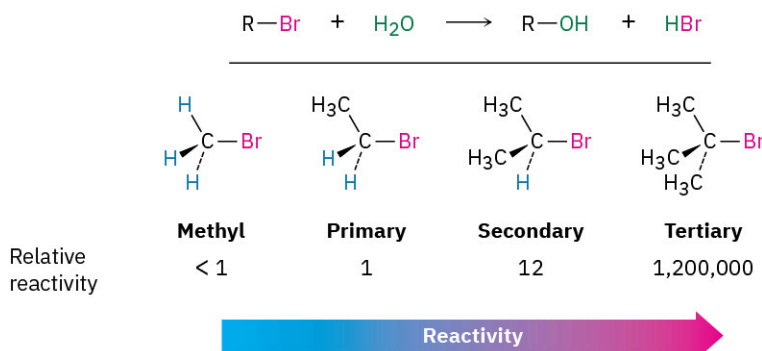


FIGURE 11.8 Energy diagrams showing the effects of (a) substrate, (b) nucleophile, (c) leaving group, and (d) solvent on S_N2 reaction rates. Substrate and leaving group effects are felt primarily in the transition state. Nucleophile and solvent effects are felt primarily in the reactant ground state.

11.4 The S_N1 Reaction

Most nucleophilic substitutions take place by the S_N2 pathway just discussed. The reaction is favored when carried out with an unhindered substrate and a negatively charged nucleophile in a polar aprotic solvent, but is disfavored when carried out with a hindered substrate and a neutral nucleophile in a protic solvent. You might therefore expect the reaction of a tertiary substrate (hindered) with water (neutral, protic) to be among the slowest of substitution reactions. Remarkably, however, the opposite is true. The reaction of the tertiary halide 2-bromo-2-methylpropane ($\text{CH}_3)_3\text{CBr}$ with H_2O to give the alcohol 2-methyl-2-propanol is more than *1 million times* faster than the corresponding reaction of CH_3Br to give methanol.



What's going on here? A nucleophilic substitution reaction is occurring—a hydroxyl group is replacing a halogen—yet the reactivity order seems backward. These reactions can't be taking place by the S_N2 mechanism we've been discussing, so we must therefore conclude that they are occurring by an alternative substitution

mechanism. This alternative mechanism is called the **S_N1 reaction**, for *substitution, nucleophilic, unimolecular*.

In contrast to the S_N2 reaction of CH₃Br with OH[−], the S_N1 reaction of (CH₃)₃CBr with H₂O has a rate that depends only on the alkyl halide concentration and is independent of the H₂O concentration. In other words, the process is a **first-order reaction**; the concentration of the nucleophile does not appear in the rate equation.

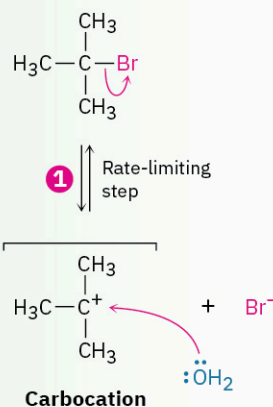
$$\begin{aligned}\text{Reaction rate} &= \text{Rate of disappearance of alkyl halide} \\ &= k \times [\text{RX}]\end{aligned}$$

To explain this result, we need to know more about kinetics measurements. Many organic reactions occur in several steps, one of which usually has a higher-energy transition state than the others and is therefore slower. We call this step with the highest transition-state energy the **rate-limiting step**, or **rate-determining step**. No reaction can proceed faster than its rate-limiting step, which acts as a kind of traffic jam, or bottleneck. In the S_N1 reaction of (CH₃)₃CBr with H₂O, the fact that the nucleophile concentration does not appear in the first-order rate equation means that it is not involved in the rate-limiting step and must therefore be involved in some other, non-rate-limiting step. The mechanism shown in **FIGURE 11.9** accounts for these observations.

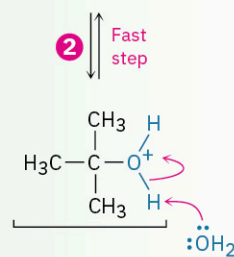
FIGURE 11.9 MECHANISM

The mechanism of the S_N1 reaction of 2-bromo-2-methylpropane with H₂O involves three steps. Step **1**—the spontaneous, unimolecular dissociation of the alkyl bromide to yield a carbocation—is rate-limiting.

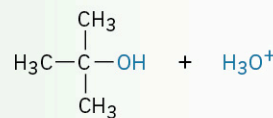
1 Spontaneous dissociation of the alkyl bromide occurs in a slow, rate-limiting step to generate a carbocation intermediate plus bromide ion.



2 The carbocation intermediate reacts with water as a nucleophile in a fast step to yield protonated alcohol as product.



3 Loss of a proton from the protonated alcohol intermediate then gives the neutral alcohol product.



Unlike what occurs in an S_N2 reaction, where the leaving group is displaced while the incoming nucleophile approaches, an S_N1 reaction takes place by loss of the leaving group *before* the nucleophile approaches. 2-Bromo-2-methylpropane spontaneously dissociates to the *tert*-butyl carbocation (CH₃)₃C⁺, plus Br[−] in a slow,

rate-limiting step, and the intermediate carbocation is then immediately trapped by the nucleophile water in a faster second step. Thus, water is not a reactant in the step whose rate is measured. The energy diagram is shown in **FIGURE 11.10**.

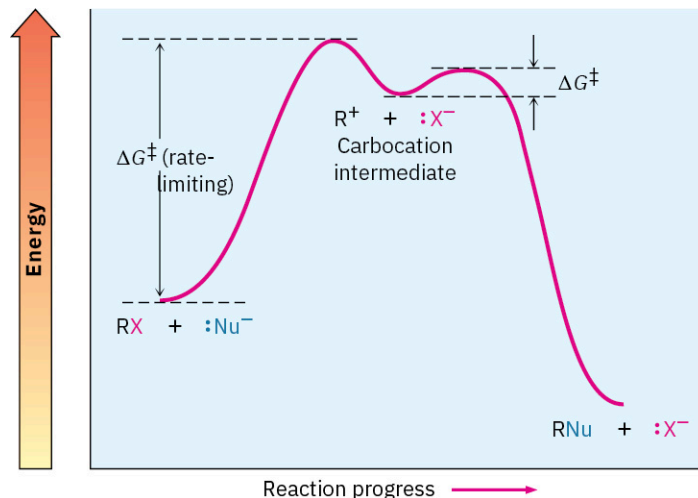


FIGURE 11.10 An energy diagram for an S_N1 reaction. The rate-limiting step is the spontaneous dissociation of the alkyl halide to give a carbocation intermediate. Reaction of the carbocation with a nucleophile then occurs in a second, faster step.

Because an S_N1 reaction occurs through a carbocation intermediate, its stereochemical outcome is different from that of an S_N2 reaction. Carbocations, as we've seen, are planar, sp^2 -hybridized, and achiral. Thus, if we carry out an S_N1 reaction on one enantiomer of a chiral reactant and go through an achiral carbocation intermediate, the product loses its optical activity (**Section 8.12**). That is, the symmetrical intermediate carbocation can react with a nucleophile equally well from either side, leading to a racemic, 50 : 50 mixture of enantiomers (**FIGURE 11.11**).

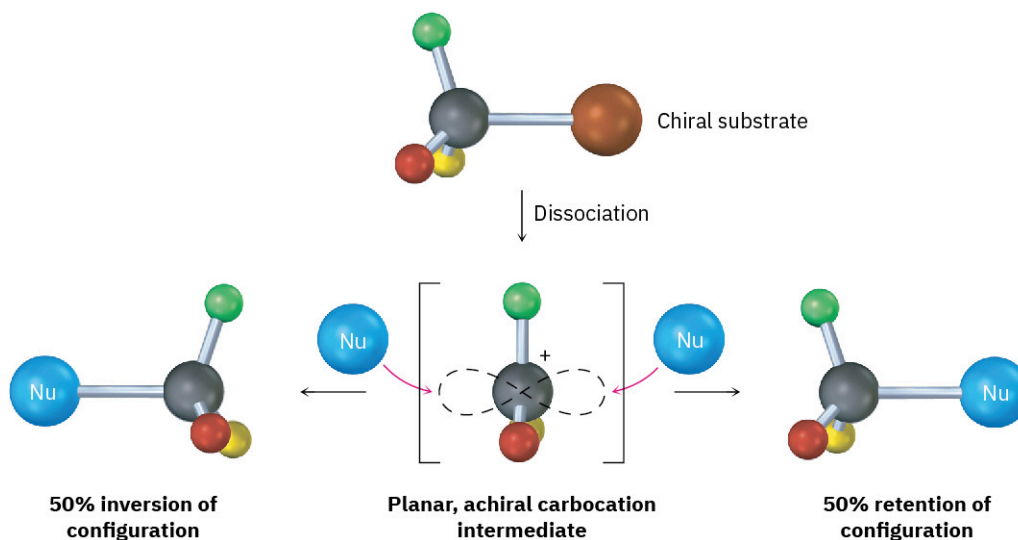
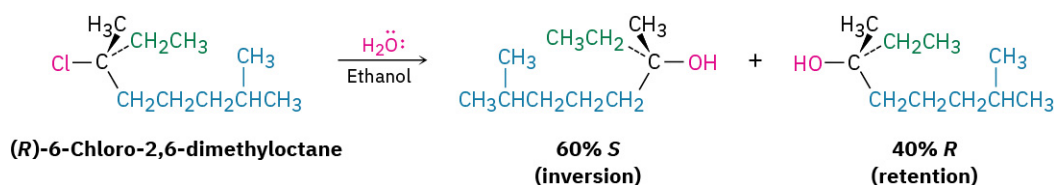


FIGURE 11.11 Stereochemistry of the S_N1 reaction. Because the reaction goes through an achiral intermediate, an enantiomerically pure reactant gives an optically inactive racemic product.

The conclusion that S_N1 reactions on enantiomerically pure substrates should give racemic products is nearly, but not exactly, what is found. In fact, few S_N1 displacements occur with complete racemization. Most give a minor (0–20%) excess of inversion. The reaction of (*R*)-6-chloro-2,6-dimethyloctane with H_2O , for example, leads to an alcohol product that is approximately 80% racemized and 20% inverted (80% *R,S* + 20% *S* is equivalent to 40% *R* + 60% *S*).



This lack of complete racemization in S_N1 reactions is due to the fact that **ion pairs** are involved. According to this explanation, first proposed by Saul Winstein at UCLA, dissociation of the substrate occurs to give a structure in which the two ions are still loosely associated and in which the carbocation is effectively shielded from reaction on one side by the departing anion. If a certain amount of substitution occurs before the two ions fully diffuse apart, then a net inversion of configuration will be observed (**FIGURE 11.12**).

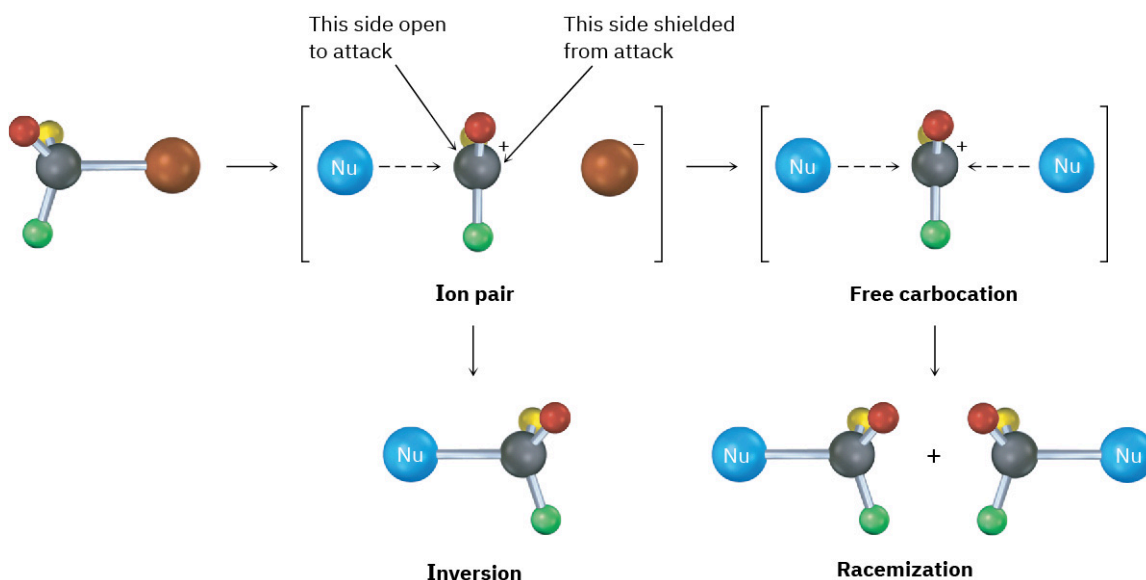


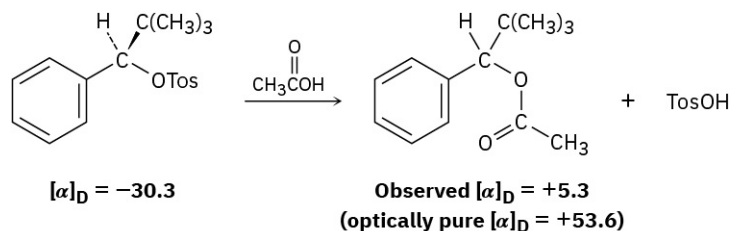
FIGURE 11.12 Ion pairs in an S_N1 reaction. The leaving group shields one side of the carbocation intermediate from reaction with the nucleophile, thereby leading to some inversion of configuration rather than complete racemization.

PROBLEM What product(s) would you expect from reaction of (*S*)-3-chloro-3-methyloctane with acetic acid?

11-8 Show the stereochemistry of both reactant and product.

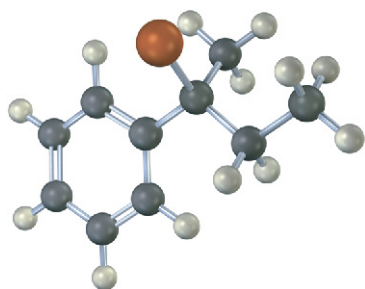
PROBLEM Among the many examples of S_N1 reactions that occur with incomplete racemization, the optically pure tosylate of 2,2-dimethyl-1-phenyl-1-propanol ($[\alpha]_D = -30.3$) gives the corresponding acetate ($[\alpha]_D = +5.3$) when heated in acetic acid. If complete inversion had occurred, the optically pure acetate would have had $[\alpha]_D = +53.6$. What percentage racemization and what percentage inversion occurred in this reaction?

11-9



PROBLEM Assign configuration to the following substrate, and show the stereochemistry and identity of the

11-10 product you would obtain by S_N1 reaction with water (reddish brown = Br):



11.5 Characteristics of the S_N1 Reaction

Just as the S_N2 reaction is strongly influenced by the structure of the substrate, the leaving group, the nucleophile, and the solvent, the S_N1 reaction is similarly influenced. Factors that lower $\Delta G^\ddagger$, either by lowering the energy level of the transition state or by raising the energy level of the ground state, favor faster S_N1 reactions. Conversely, factors that raise $\Delta G^\ddagger$, either by raising the energy level of the transition state or by lowering the energy level of the reactant, slow down the S_N1 reaction.

The Substrate

According to the Hammond postulate (Section 7.10), any factor that stabilizes a high-energy intermediate also stabilizes the transition state leading to that intermediate. Because the rate-limiting step in an S_N1 reaction is the spontaneous, unimolecular dissociation of the substrate to yield a carbocation, the reaction is favored whenever a stabilized carbocation intermediate is formed. The more stable the carbocation intermediate, the faster the S_N1 reaction.

We saw in Section 7.9 that the stability order of alkyl carbocations is $3^\circ > 2^\circ > 1^\circ > \text{methyl}$. To this list we should also add the resonance-stabilized allyl and benzylic cations. Just as allylic radicals are unusually stable because the unpaired electron can be delocalized over an extended π orbital system (Section 10.4), so allylic and benzylic carbocations are unusually stable. (The word **benzylic** means “next to an aromatic ring.”) As FIGURE 11.13 indicates, an allylic cation has two resonance forms. In one form, the double bond is on the “left”; in the other form it’s on the “right.” A benzylic cation has five resonance forms, all of which contribute to the overall resonance hybrid.

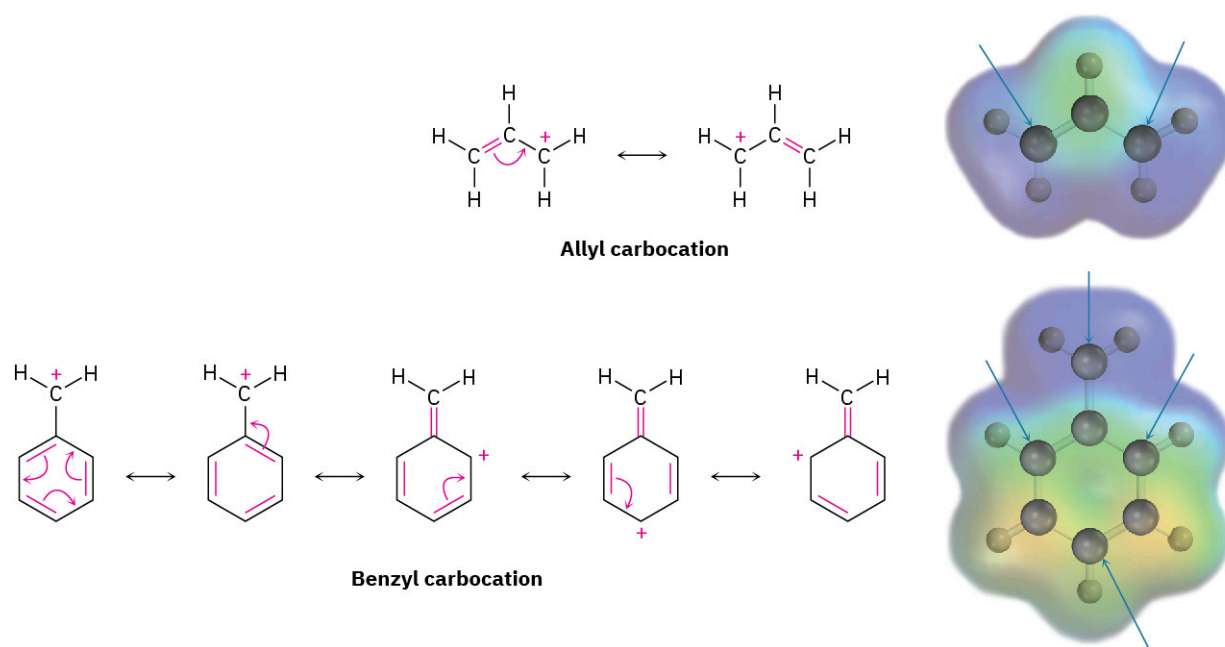
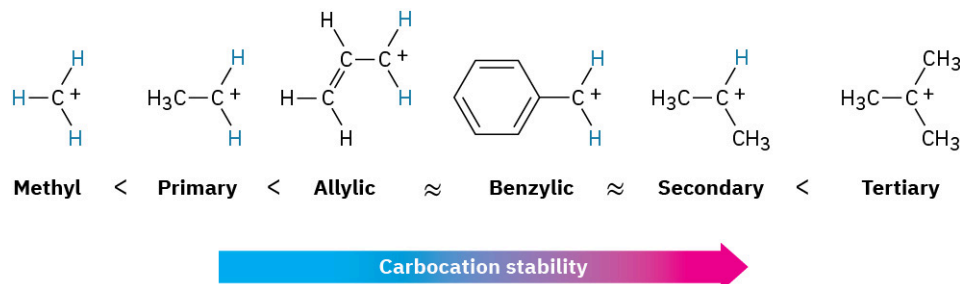


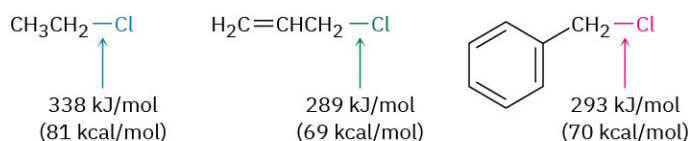
FIGURE 11.13 Resonance forms of allylic and benzylic carbocations. The positive charge is delocalized over the π system in both. **Electron-poor atoms** are indicated by blue arrows.

Because of resonance stabilization, a primary allylic or benzylic carbocation is about as stable as a secondary

alkyl carbocation, and a secondary allylic or benzylic carbocation is about as stable as a tertiary alkyl carbocation. This stability order of carbocations is the same as the order of S_N1 reactivity for alkyl halides and tosylates.

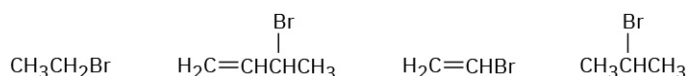


We should also note parenthetically that primary allylic and benzylic substrates are particularly reactive in S_N2 reactions as well as in S_N1 reactions. Allylic and benzylic C–X bonds are about 50 kJ/mol (12 kcal/mol) weaker than the corresponding saturated bonds and are therefore more easily broken.



PROBLEM Rank the following substances in order of their expected S_N1 reactivity:

11-11



PROBLEM 3-Bromo-1-butene and 1-bromo-2-butene undergo S_N1 reaction at nearly the same rate, even though one is a secondary halide and the other is primary. Explain.

11-12

The Leaving Group

We said during the discussion of S_N2 reactivity that the best leaving groups are those that are most stable; that is, those that are the conjugate bases of strong acids. An identical reactivity order is found for the S_N1 reaction because the leaving group is directly involved in the rate-limiting step. Thus, the S_N1 reactivity order is



Note that in the S_N1 reaction, which is often carried out under acidic conditions, neutral water is sometimes the leaving group. This occurs, for example, when an alkyl halide is prepared from a tertiary alcohol by reaction with HBr or HCl (Section 10.5). As shown in Figure 11.14, the alcohol is first protonated and then spontaneously loses H₂O to generate a carbocation, which reacts with halide ion to give the alkyl halide. Knowing that an S_N1 reaction is involved in the conversion of alcohols to alkyl halides explains why the reaction works well only for tertiary alcohols. Tertiary alcohols react fastest because they give the most stable carbocation intermediates.

FIGURE 11.14 MECHANISM

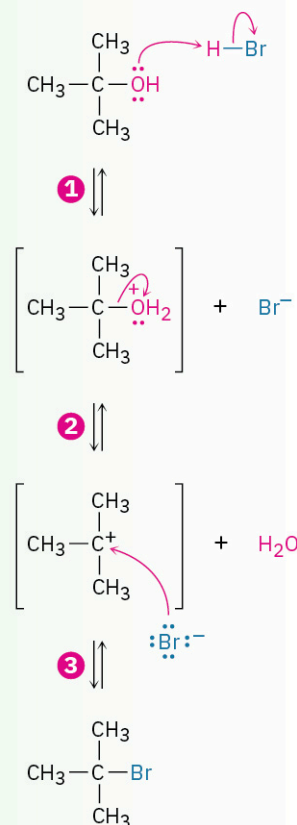
The mechanism of the S_N1 reaction of a tertiary alcohol with HBr to yield an alkyl halide. Neutral water is the leaving group (step 2).

1 The -OH group is first protonated by HBr.

2 Spontaneous dissociation of the protonated alcohol occurs in a slow, rate-limiting step to yield a carbocation intermediate plus water.

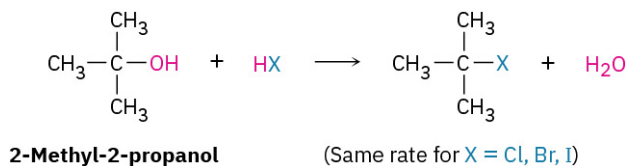
Carbocation

3 The carbocation intermediate reacts with bromide ion in a fast step to yield the neutral substitution product.



The Nucleophile

The nature of the nucleophile plays a major role in the S_N2 reaction but does not affect an S_N1 reaction. Because the S_N1 reaction occurs through a rate-limiting step in which the added nucleophile has no part, the nucleophile can't affect the reaction rate. The reaction of 2-methyl-2-propanol with HX, for instance, occurs at the same rate regardless of whether X is Cl, Br, or I. Furthermore, neutral nucleophiles are just as effective as negatively charged ones, so S_N1 reactions frequently occur under neutral or acidic conditions.



The Solvent

What about the solvent? Do solvents have the same effect in S_N1 reactions that they have in S_N2 reactions? The answer is both yes and no. Yes, solvents have a large effect on S_N1 reactions, but no, the reasons for the effects on S_N1 and S_N2 reactions are not the same. Solvent effects in the S_N2 reaction are due largely to stabilization or destabilization of the nucleophile *reactant*, while solvent effects in the S_N1 reaction are due largely to stabilization or destabilization of the *transition state*.

The Hammond postulate says that any factor stabilizing the intermediate carbocation should increase the rate

of an S_N1 reaction. Solvation of the carbocation—the interaction of the ion with solvent molecules—has such an effect. Solvent molecules orient around the carbocation so that the electron-rich ends of the solvent dipoles face the positive charge (**FIGURE 11.15**), thereby lowering the energy of the ion and favoring its formation.

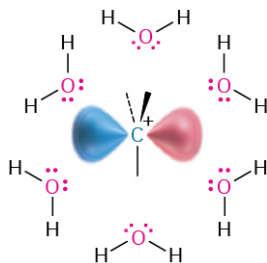
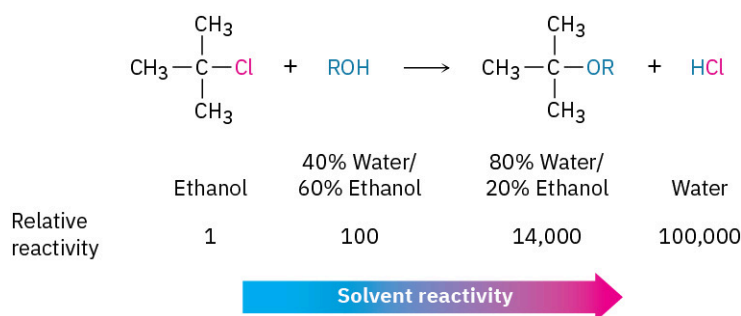


FIGURE 11.15 Solvation of a carbocation by water. The electron-rich oxygen atoms of solvent molecules orient around the positively charged carbocation and thereby stabilize it.

The properties of a solvent that contribute to its ability to stabilize ions by solvation are related to the solvent's polarity. S_N1 reactions take place much more rapidly in strongly polar solvents, such as water and methanol, than in less polar solvents, such as ether and chloroform. In the reaction of 2-chloro-2-methylpropane, for example, a rate increase of 100,000 is observed upon going from ethanol (less polar) to water (more polar). The rate increases when going from a hydrocarbon solvent to water are so large they can't be measured accurately.



It should be emphasized again that both the S_N1 and the S_N2 reaction show solvent effects, but that they do so for different reasons. S_N2 reactions are *disfavored* in protic solvents because the *ground-state energy* of the nucleophile is lowered by solvation. S_N1 reactions are *favored* in protic solvents because the *transition-state energy* leading to carbocation intermediate is lowered by solvation.

A Summary of S_N1 Reaction Characteristics

The effects on S_N1 reactions of the four variables—substrate, leaving group, nucleophile, and solvent—are summarized in the following statements:

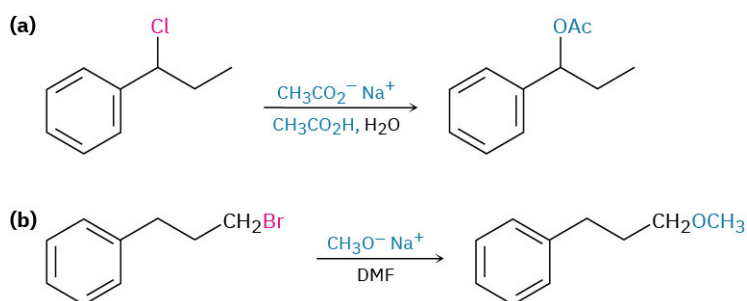
Substrate	The best substrates yield the most stable carbocations. As a result, S _N 1 reactions are best for tertiary, allylic, and benzylic halides.
Leaving group	Good leaving groups increase the reaction rate by lowering the energy level of the transition state for carbocation formation.
Nucleophile	The nucleophile must be nonbasic to prevent a competitive elimination of HX (Section 11.7), but otherwise does not affect the reaction rate. Neutral nucleophiles work well.
Solvent	Polar solvents stabilize the carbocation intermediate by solvation, thereby increasing the reaction rate.



WORKED EXAMPLE 11.2

Predicting the Mechanism of a Nucleophilic Substitution Reaction

Predict whether each of the following substitution reactions is likely to be S_N1 or S_N2:

**Strategy**

Look at the substrate, leaving group, nucleophile, and solvent. Then decide from the summaries at the ends of [Section 11.3](#) and [Section 11.5](#) whether an S_N1 or an S_N2 reaction is favored. S_N1 reactions are favored by tertiary, allylic, or benzylic substrates, by good leaving groups, by nonbasic nucleophiles, and by protic solvents. S_N2 reactions are favored by primary substrates, by good leaving groups, by good nucleophiles, and by polar aprotic solvents.

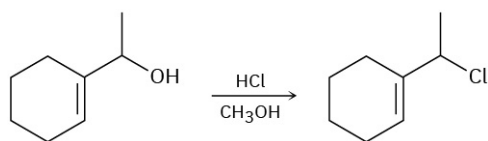
Solution

(a) This is likely to be an S_N1 reaction because the substrate is secondary and benzylic, the nucleophile is weakly basic, and the solvent is protic.

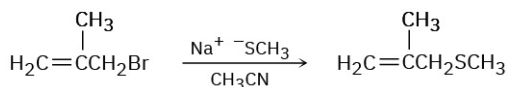
(b) This is likely to be an S_N2 reaction because the substrate is primary, the nucleophile is a good one, and the solvent is polar aprotic.

PROBLEM Predict whether each of the following substitution reactions is likely to be S_N1 or S_N2 :

11-13 (a)



(b)



11.6 Biological Substitution Reactions

Both S_N1 and S_N2 reactions are common in biological chemistry, particularly in the pathways for biosynthesis of the many thousands of plant-derived substances called terpenoids, which we saw briefly in the [Chapter 8 Chemistry Matters](#) and will discuss in [Section 27.5](#). Unlike what typically happens in the laboratory, however, the substrate in a biological substitution reaction is usually an organodiphosphate rather than an alkyl halide. Thus, the leaving group is the diphosphate ion, abbreviated PP_i , rather than a halide ion. In fact, it's useful to think of the diphosphate group as the “biological equivalent” of a halide. The dissociation of an organodiphosphate in a biological reaction is typically assisted by complexation to a divalent metal cation such as Mg^{2+} to help neutralize charge and make the diphosphate a better leaving group.